

Gheat

Systems Engineer · Quantitative Developer · gheatmc@gmail.com · gheat.net

SUMMARY

Highly technical software engineer, systems administrator, and quantitative developer with deep experience in full-stack development, AI infrastructure, cryptography, and high-performance bare-metal networking. Proven track record architecting enterprise-grade pipelines, processing multi-dimensional vector databases, and automating bare-metal infrastructure deployments. Specialized in cyber security, advanced mathematical modeling, and containerized cloud/edge environments.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C#, C, C++, PHP, SQL, HTML5, CSS3, Bash, Rust

AI / ML: Retrieval-Augmented Generation (RAG), High-Dimensional Vector Databases, MCP Tool Creation, LLM Fine-Tuning, Deep-Learning Stem Separation (Demucs), OCR Pipelines, Local AI Agents & Tool Calling (Ollama XML Agent Loops)

Security / Infrastructure: Penetration Testing & Vulnerability Remediation, XSS Structural Fixes, SQL Injection Mitigation, Directory Traversal Defense, JWT Secret Forgery Prevention, Network Isolation, Linux System Administration (Arch Linux, Debian), L2/L3 Network Switching, Tailscale Mesh Networking, Cloudflare Tunnels, Reverse Proxies (Caddy), Docker Containerization

Hardware / Engineering: 19" Data Center Rack Architecture, Bare-Metal Server Provisioning, Additive Manufacturing (3D Printing), Hardware Modding, Systems Engineering

EXPERIENCE

Hedge Fund — Quantitative Software Engineering Intern

Summer 2025

- Selected as the first-ever paid technical intern in the firm's history based on advanced project scope, technical execution, and mathematical aptitude.
- Architected and deployed a production-grade automated data ingestion pipeline using Python and advanced OCR tools to extract unstructured financial data from legacy handwritten contracts.
- Engineered a high-dimensional vector database solution using Python and PHP to convert extracted textual financial data into searchable mathematical embeddings for quantitative analysis.
- Implemented secure server-side logic and optimized data flow using structured SQL queries to securely cross-reference internal transactional ledger records.
- Applied deep financial insights and interest in cryptocurrency markets to optimize automated legal and data workflows for digital asset analysis.

Independent DevOps & Game Development — Lead Systems Engineer & Security Analyst

2024 – Present

- Engineered, deployed, and maintain a private bare-metal data center environment utilizing an enterprise-grade 19-inch equipment rack to host high-performance AI workloads, dynamic game servers, and full-stack web applications.
- Audited application security across self-hosted titles, identifying and neutralizing critical vulnerabilities including database-extracting SQL injections, file structure traversal attacks, JWT secret forgery, hit fraud, asset-farming exploits, XSS vectors, and IP leaks.
- Provision, secure, and maintain custom headless Arch Linux spin-ups across all bare-metal nodes, writing custom bash automation scripts for rapid server deployment, security hardening, and performance monitoring.
- Built and styled full-stack web applications and custom dashboards using modern HTML, CSS, and JavaScript.
- Combined advanced mathematical problem-solving with custom 3D printing and hardware modifications to design optimal cooling and physical rack layouts for continuous high-compute AI operations.

- Volunteered time teaching computer science, programming fundamentals, and technology literacy to younger students, breaking down complex coding concepts into easy-to-understand foundational lessons.

PROJECTS

Atlas

AI-Driven Desktop Control Panel & Agentic OS

- Architected a desktop assistant environment with a React 19 / TypeScript frontend and a Rust backend powered by Tauri 2. Built a persistent local AI agent loop (Ollama) that parses system state and executes system-level operations via an XML-style tag emission engine. Engineered a native terminal with a real SSH shell (xterm.js), a Rust PTY bridge, and SFTP browsing; integrated markdown note sync, an IMAP/SMTP email client with AI drafting, calendar/task control, and subscription ledger metrics. Secured all node communication over a Tailscale mesh with no public network exposure.

Wavly

High-Precision DJ Music Similarity Engine

- Built a local music matching engine that isolates kick drums and basslines via deep-learning stem separation (Demucs) to compute independent similarity scores. Engineered a feature extraction pipeline (librosa, numpy, scipy) producing high-dimensional vectors; ranks 1000+ tracks in milliseconds using cached JSON feature vectors and L2-normalised cosine distance. Deployed on LAN behind a waitress WSGI server with CUDA-accelerated torch builds and strict private-IP network isolation.

Spaceships

Real-Time Multiplayer 3D Combat Game

- Architected the online backend and deployment pipelines for a 3D space combat game live at gheat.net/spaceships. Engineered low-latency WebSocket communication (Node.js, Express) for real-time multiplayer sync and combat events; persistent data tier on SQLite (better-sqlite3) with bcrypt-hashed credentials and JWT auth. Collaborated on client optimizations — message batching, frustum culling, particle tracking — to hold stable 60 FPS.

Voidwatch

Space Exploration Simulation Game

- Designed and built a scalable space exploration game in C#, using object-oriented design to manage complex physics, mathematical simulations, and real-time state synchronization.

SimplyServer Dashboard

Python · Tailscale API · Linux Administration

- Developed a cross-platform desktop GUI application in Python to manage, monitor, and configure headless remote servers securely over encrypted Tailscale mesh networks.

Custom MCP Tools & AI Automation

Python · LLMs · Developer Workflows

- Built and deployed custom model context protocol (MCP) tools interfacing local hardware and databases with large language models, streamlining developer workflow automation and model training pipelines.